

Robust Plant Disease Classification Under Real-World Distribution Shifts

Team 81: Luisa Chavez, Chase Wrenn, Joyce Zhou

Problem Statement

- **Motivation:**
 - Major cause of crop yield loss
 - Early detection is critical
 - Automated diagnosis is scalable
- **Task:** multi-class classification
- **Input:** RGB image of a leaf
- **Output:** disease category (probability over classes)



20-40% global
yield loss [1]

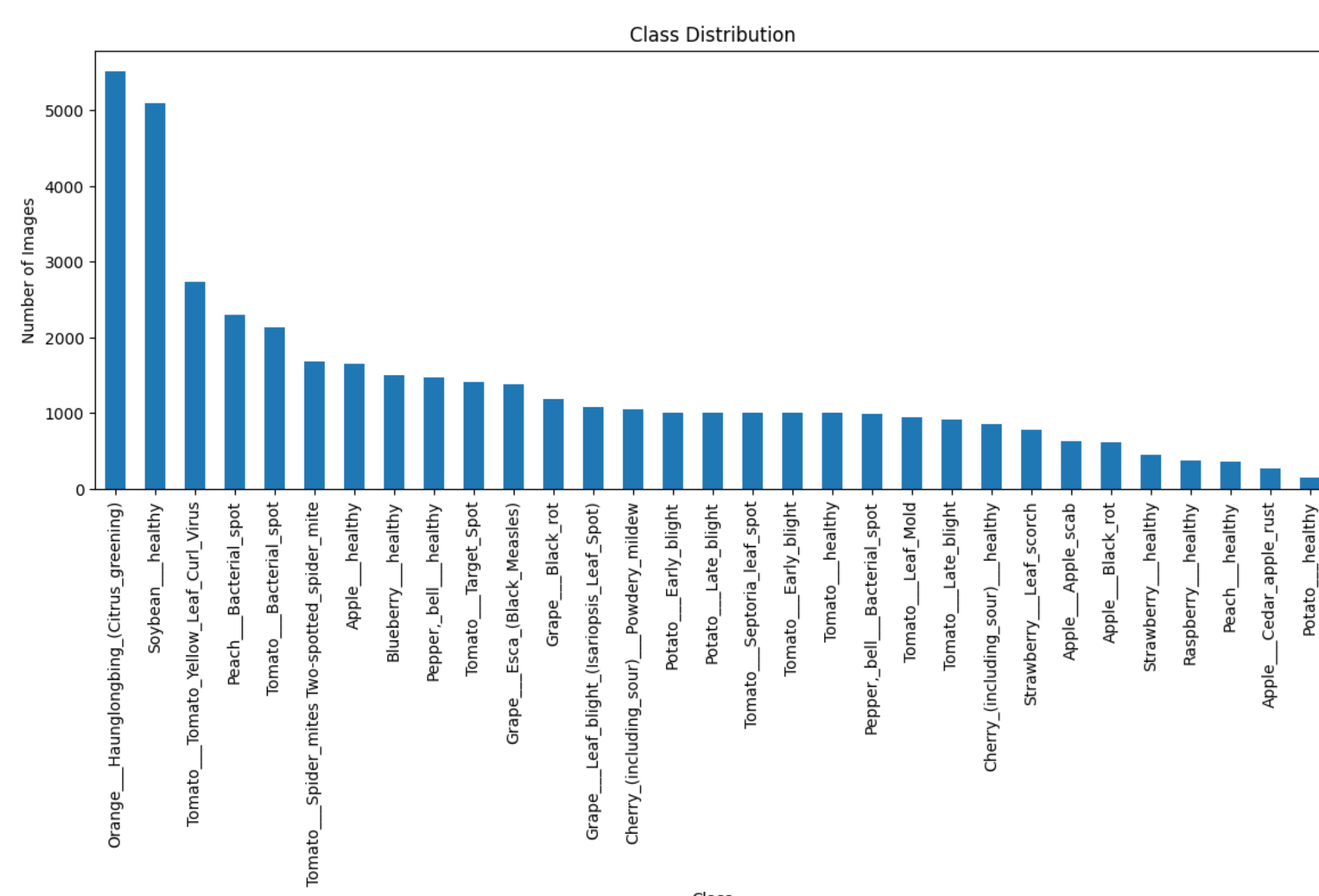
\$220B+ annual
economic losses
globally [1]

Data

Plant Village Dataset [2]
~ 50k images across 38 classes

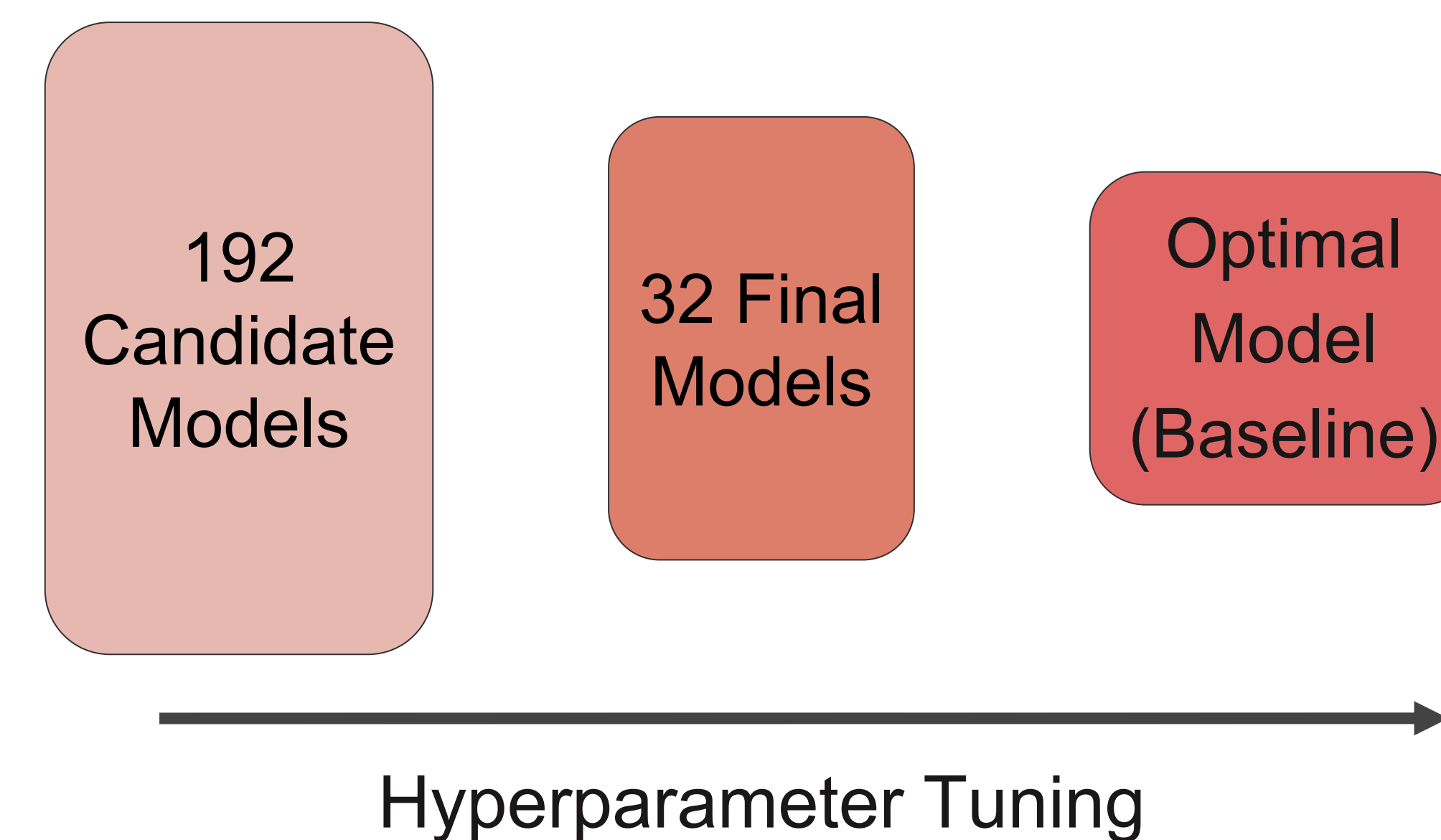
- Built metadata from raw images
- Introduced **leaf_id** to group images from the same leaf
- Leaf-level split
 - Prevents data leakage
 - Avoids nearly identical images across splits

After Cleaning: ~42k images, 8k leaves, 31 classes
Notice some Class imbalance → accuracy not sufficient

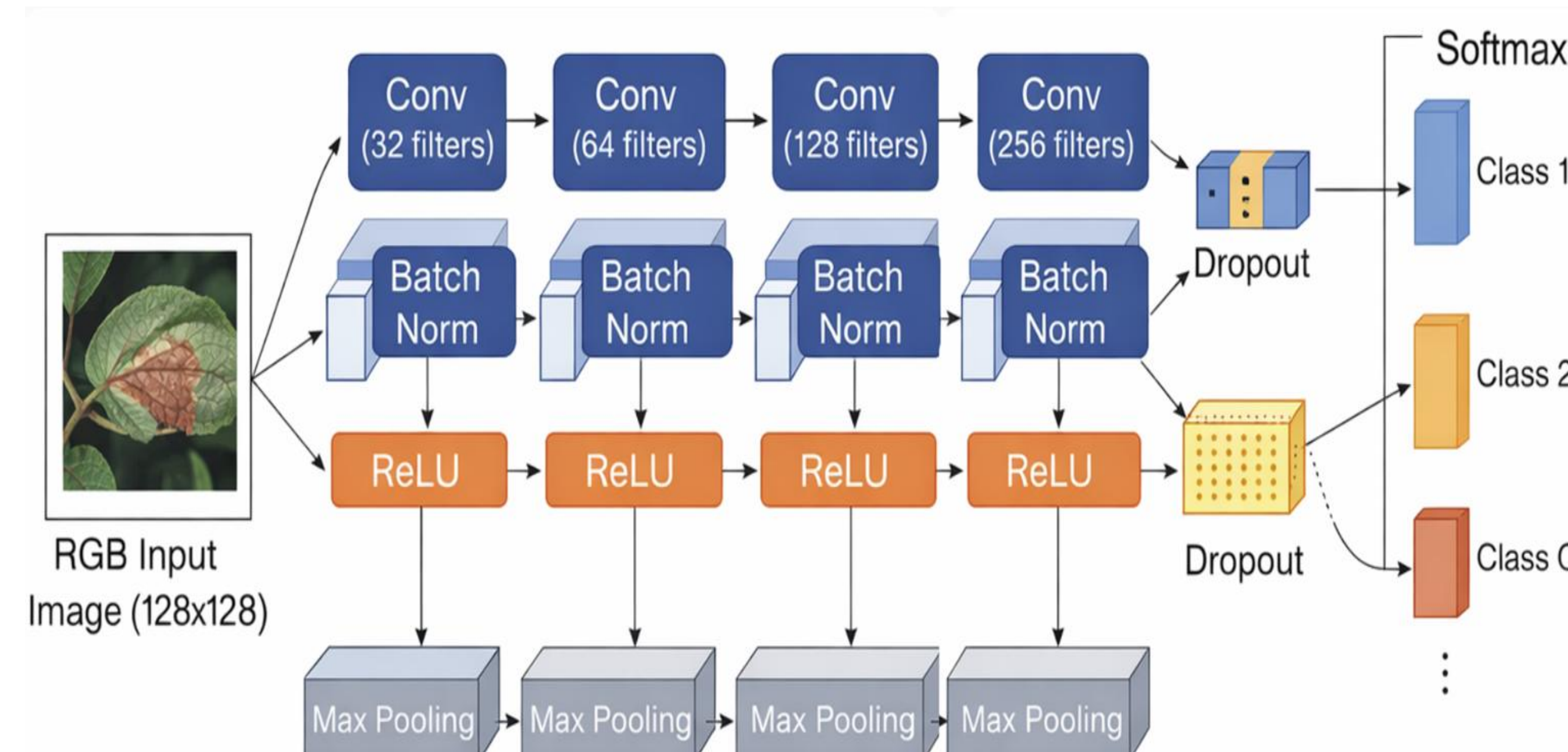


Baseline: Logistic Regression

- **Hyperparameter Choice:**
 - Regularization
 - Convergence
 - Solver Type
- **Test Process:** 192 Initial Candidates, 32 Final Candidates
- **Best Candidate Accuracy:** ~63.65%
- **Optimal Parameters:** SAGA solver, 0.05 convergence tolerance, regularization strength of 1



CNN Model



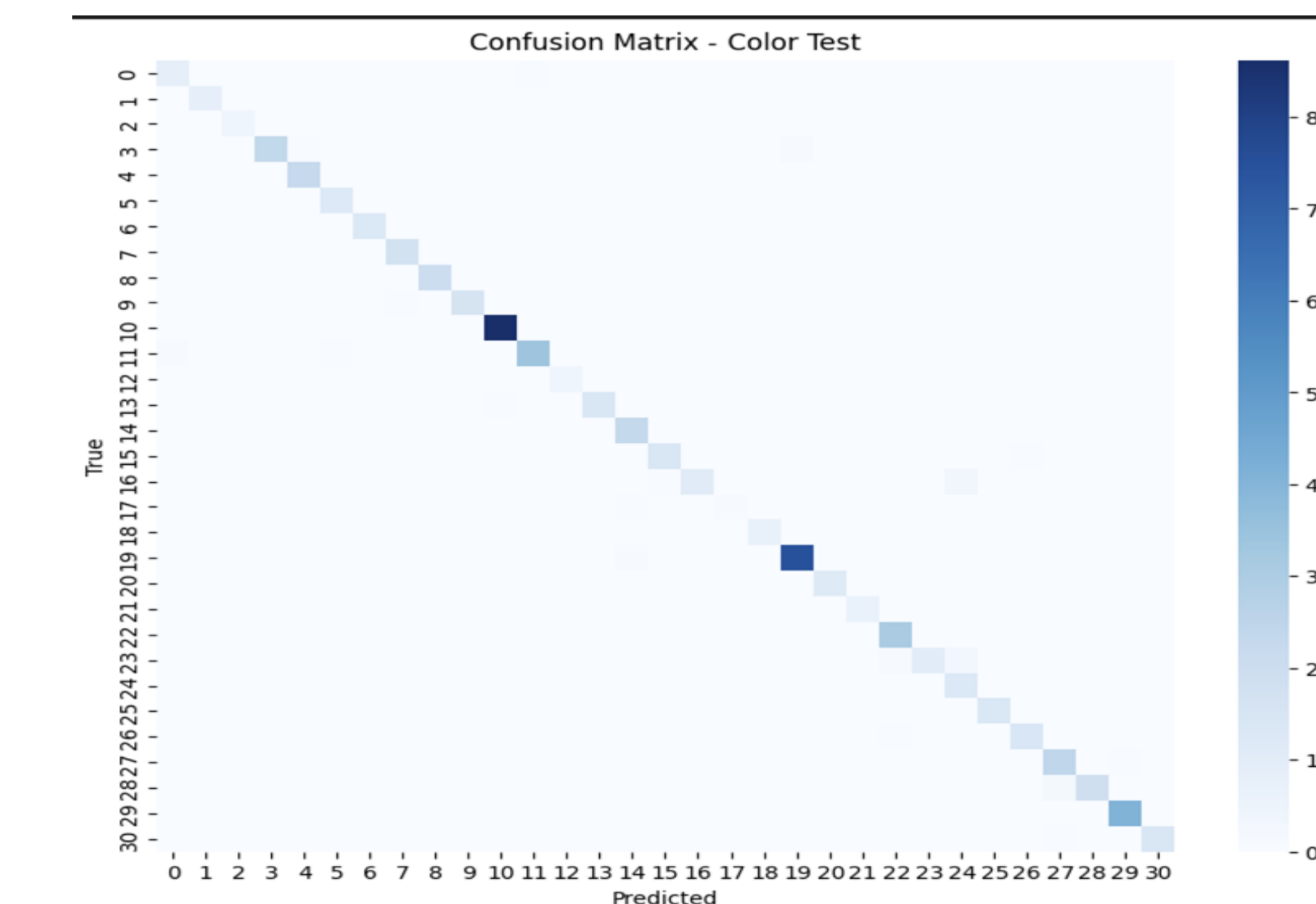
- Designed to balance model capacity vs generalization
- Custom CNN with 4 convolutional blocks
 - Filters: 32 → 64 → 128 → 256
 - Each block: Conv + BatchNorm + ReLU + MaxPool
 - Dropout used in fully connected layers
- Input size: 128×128 RGB images
- Output: Softmax over 31 classes

Evaluation Results

Color Test (Clean Data):

Accuracy: 0.96 | Macro-F1: 0.94

Confusion Matrix - Color Test

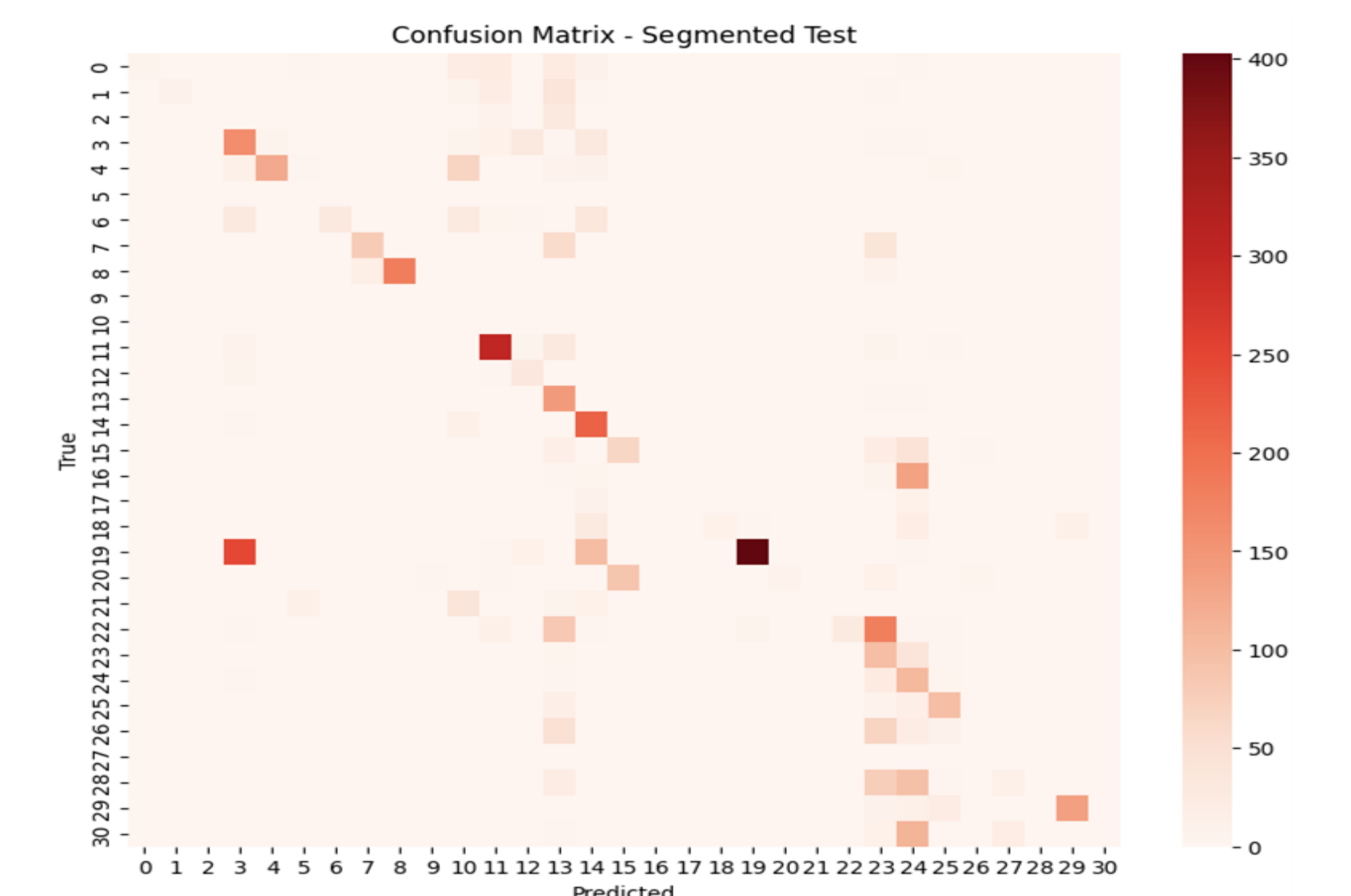


- High performance on clean data.

Segmented Test (Shifted Data):

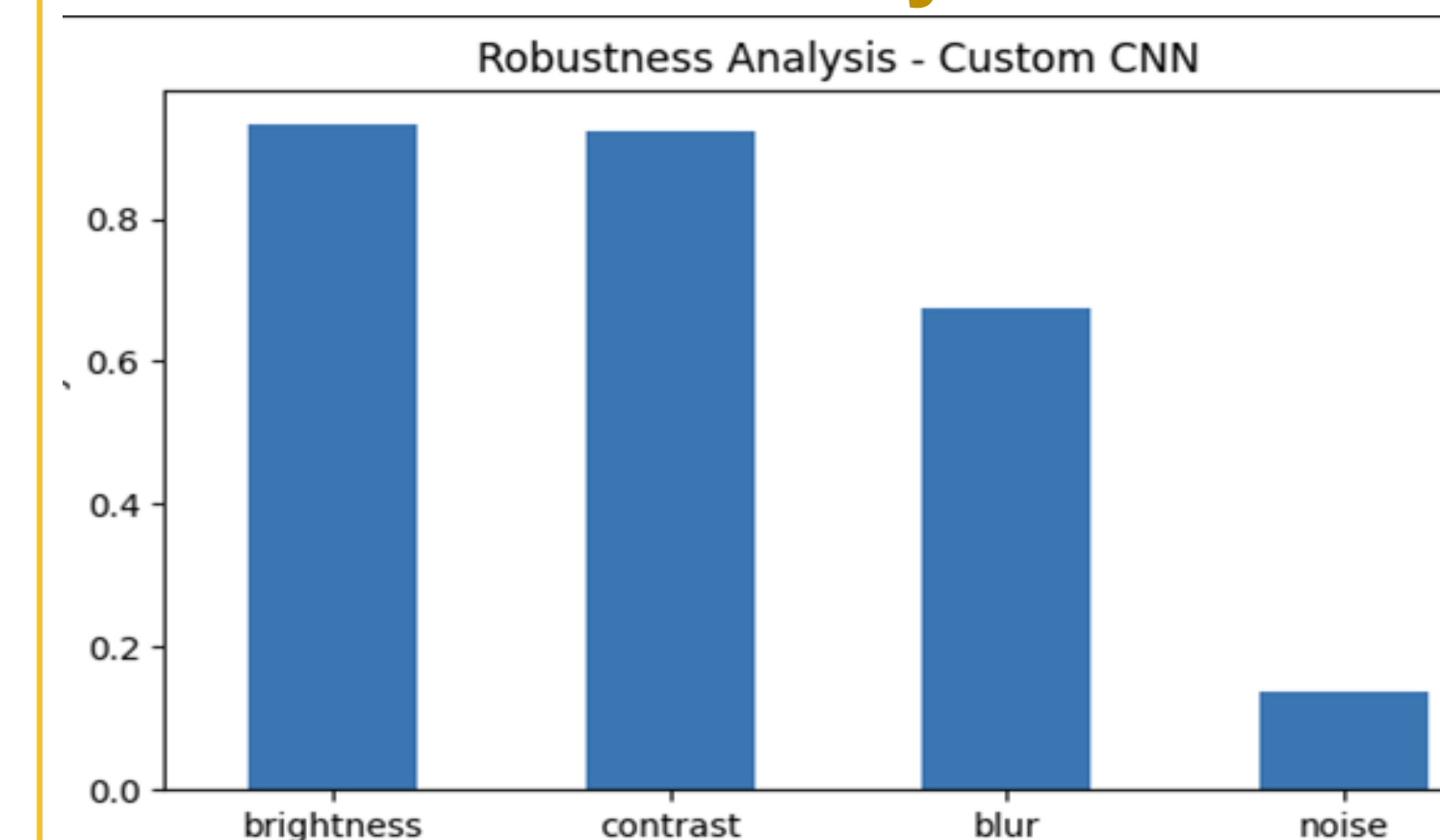
Accuracy: 0.47 | Macro-F1: 0.30

Confusion Matrix - Segmented Test



- Severe drop under distribution shift

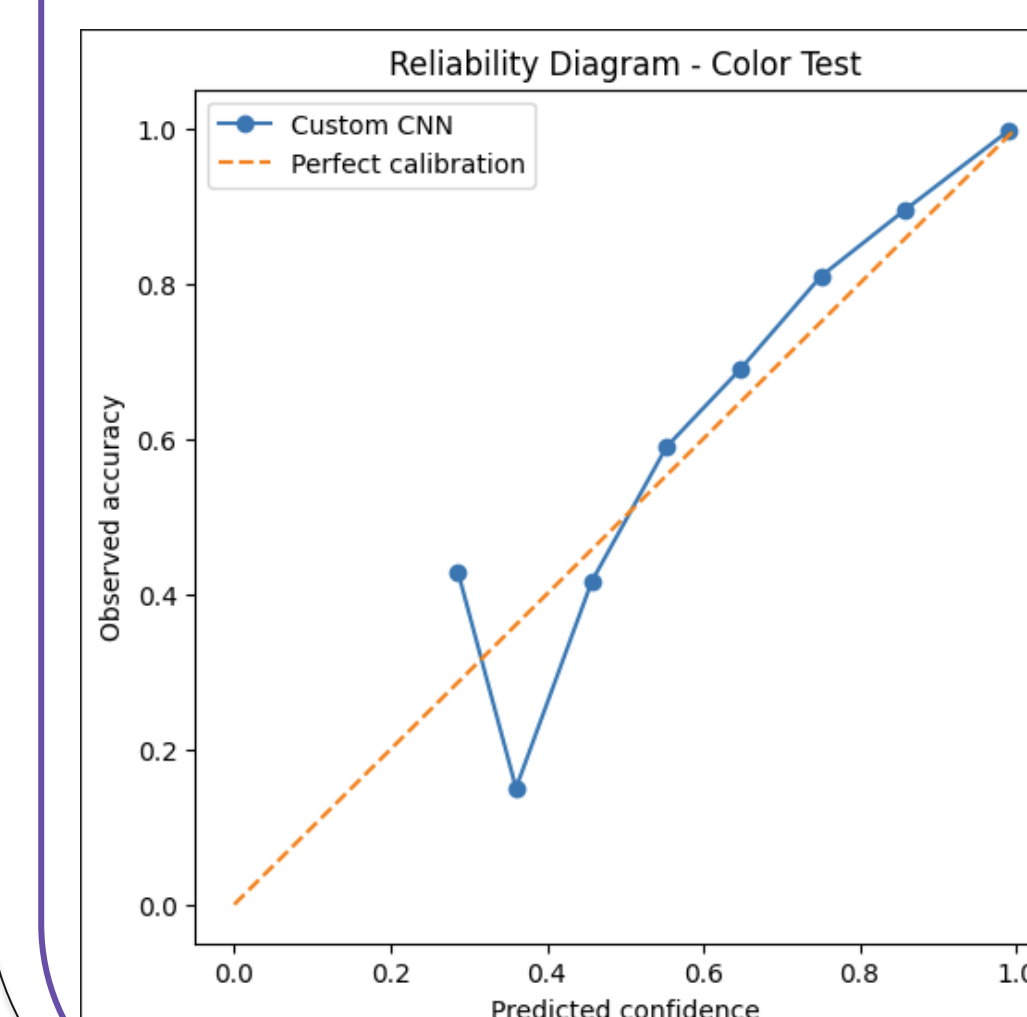
Robustness Analysis:



	Accuracy	Macro-F1
brightness	0.9322	0.9093
contrast	0.9234	0.8939
blur	0.6744	0.5919
noise	0.1379	0.1112

- Robust to brightness & contrast
- Degrades under blur
- Severe drop with noise

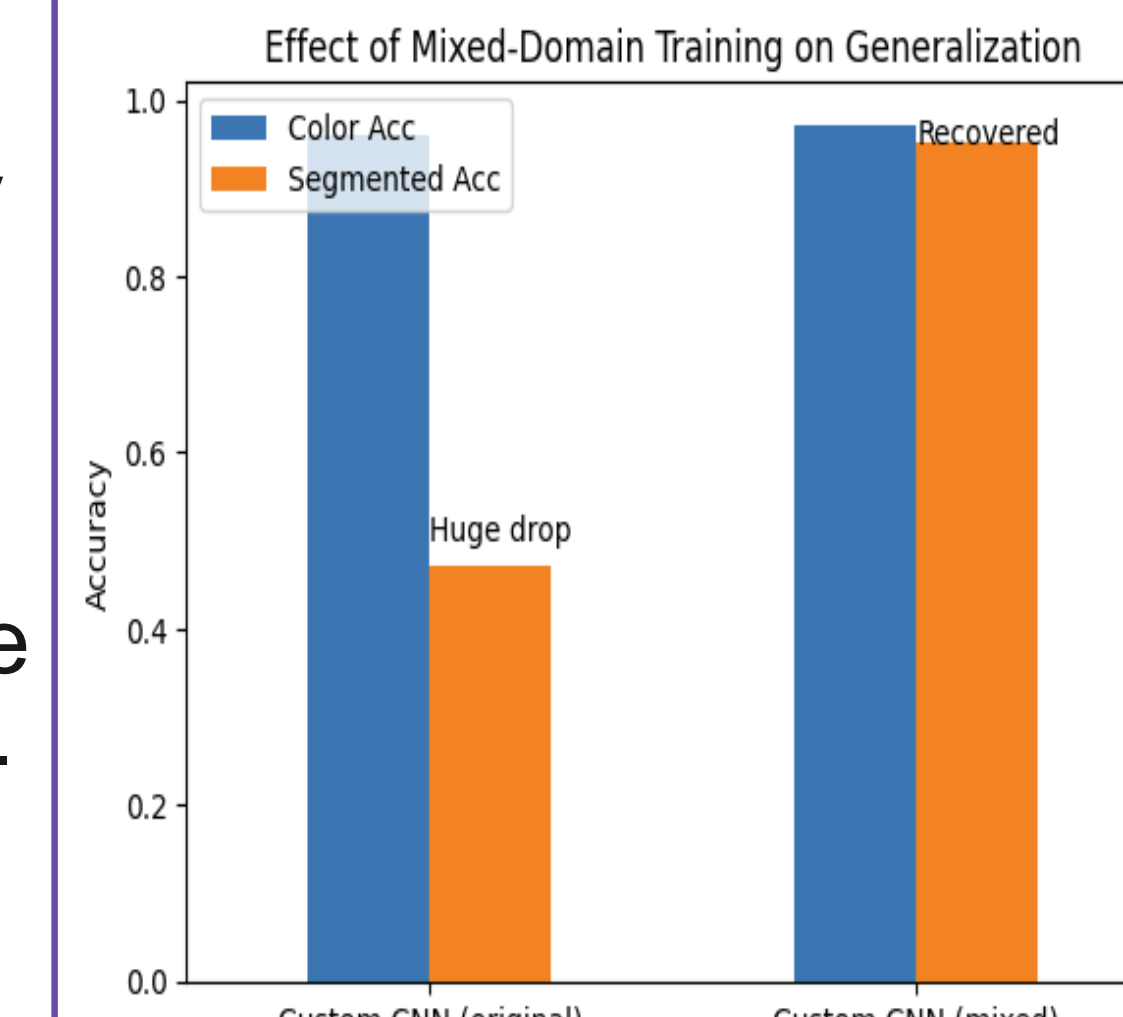
Calibration (Reliability)



- Model is reasonably calibrated.

- Slight overconfidence at mid-confidence

Generalization Improvement:



- Mixed-domain training recovers performance under distribution shift.
- Strongly reduces the generalization gap

Conclusions

- Models trained on clean data fail under distribution shifts, revealing a significant generalization gap.
- Mixed-domain training significantly improves performance, showing that data diversity is key for real-world robustness.
- Robustness evaluation is essential, as benchmark accuracy alone can be misleading in real-world settings.

Main References

1. FAO. Plant Production and Protection Division, 2021. <https://www.fao.org/plant-production-protection/>
2. PlantVillage Dataset, Mohanty et al., 2016. <https://github.com/spmohanty/PlantVillage-Dataset>
3. Crop leaf images from real field conditions. ResearchGate. <https://www.researchgate.net/>